



GENERAL INTRODUCTION

Organizational practices are now dependent not only on human resources but also on the ability to manage processes effectively and adapt to change due to the rapid evolution of digital technologies. In this context, recruiting has a strategic importance as it directly impacts team building and organizational success. However, the complexity of this process has increased because of the growing number of applications and the demand for faster, more accurate treatment. Consequently, organizations are turning to digital solutions that centralize information, improve workflows, and support decision-making, turning recruitment into a structured process that requires efficient tools.

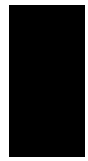
In this setting, Sopra HR Software is recognized as a leading player in human resource management solutions. Operating in a field where technological innovation and service quality are essential, the company emphasizes the need to improve its internal processes and business tools. In this sphere, the digitalization of recruitment is a major lever for streamlining application processing, enhancing the user experience, and supporting recruiters in their day-to-day activities.

The implementation of an intelligent recruitment portal aims to meet both operational and strategic needs by providing a more cohesive framework for managing job offers, applications, and interactions among all stakeholders involved in recruitment. Consequently, a central question arises : how can a recruitment portal be designed to centralize applications, facilitate CV processing, and effectively support recruiters in their decision-making, while aligning with a logic of continuous improvement in the recruitment process ?

To explore the identified problem and present the work conducted during the internship, the report is divided into three main chapters :

- **Chapter One** introduces the general context of the project. It introduces the host company, outlines the project background, identifies the problem, analyzes the current system, and outlines the proposed solution. It further emphasizes the methodological and technological choices adopted for the development of the intelligent recruitment portal.
- **Chapter Two** addresses requirements analysis and system design. It describes the functional and non-functional requirements, highlights the expected features, and introduces the main design components used to build the portal.
- **Chapters Three, Four, Five, and Six** are devoted to the development and implementation phase. They outline the steps involved in building the application, detail the implemented features, and summarize the main outcomes achieved.

Finally, the report concludes with a summary that recaps the context of the work, reviews the project, and presents possible directions for future enhancements.



CHAPTER 1 : GENERAL PROJECT CONTEXT

Introduction

This chapter situates the project in its overall context and introduces the host company, Sopra HR Software, and the scope of work. It begins with a presentation of the host organization, followed by an analysis of the existing system and a critical review of its shortcomings. It then identifies the problem, outlines the proposed solution, and ends with a description of the adopted methodology for managing the project.

1. Project Context

1.1 Presentation of the Organization

Sopra Steria is a well-known European group recognized for its expertise in consulting, digital services, and business software publishing, particularly in human resources and banking. In 2024, the Sopra Steria Group operates in nearly 30 countries and employs around 51,000 people. Sopra HR Software is a subsidiary of the Sopra Steria Group, specializing in Human Resources management, payroll administration, and talent management, both nationally and internationally. More than 850 clients use its solutions across over 54 countries. Designed to meet the specific needs of HR departments, these solutions are intended for large and medium-sized enterprises in both the public and private sectors.



FIGURE 1 – Logo of Sopra HR Software Company

1.2 Problem Statement

Recruitment has become a critical and complex function for modern organizations, particularly in environments characterized by high competition and rapid growth. Human resources departments are required to attract qualified candidates, manage large volumes of applications, and make accurate hiring decisions within limited timeframes. However, traditional recruitment methods continue to present several major challenges, including :

- The increasing number of applications received for each job opening
- The manual and time-consuming processing of CVs and candidate information
- The difficulty of efficiently matching candidates' skills with job requirements
- The lack of centralized platforms to manage job offers, applications, and recruiter interactions

According to studies on electronic recruitment tools, manual recruitment processes are often repetitive, error-prone, and inefficient, especially when recruiters face large-scale CV screening and multiple recruitment stages. These constraints make it difficult for HR professionals to maintain responsiveness, objectivity, and consistency in hiring decisions.

In the context of Sopra HR Software, which operates in a sector where technological innovation and service quality are essential, these challenges highlight the need for a more structured, automated, and intelligent recruitment approach. The absence of a unified and intelligent recruitment portal can lead to fragmented information, longer hiring cycles, and reduced efficiency for recruiters.

1.3 Justification

This project is part of a broader digital transformation of human resources practices, aligned with the growing adoption of electronic recruitment (e-recruitment) and applicant tracking

systems (ATS). Research shows that online recruitment tools significantly simplify the hiring process by centralizing candidate data, automating CV sorting, and supporting decision-making through structured workflows.

The implementation of an intelligent recruitment portal goes beyond basic digitalization ; it aims to optimize the entire recruitment lifecycle. By relying on automation and centralized data management, the portal helps reduce recruitment time, minimize errors, and improve the overall efficiency of HR teams. This approach directly supports Sopra HR Software's objective of continuously improving its internal processes and delivering high-quality HR solutions to its clients.

2. Existing System Analysis

The current recruitment practices in many organizations, including enterprise environments like Sopra HR Software, often rely on multiple disconnected tools and manual operations. Job offers may be published across different platforms, while applications are received through emails, job boards, or internal systems. Recruiters must manually review CVs, shortlist candidates, coordinate interviews, and track feedback.

As highlighted in studies on online recruitment systems, handling large volumes of applications without an automated applicant tracking system makes it difficult to identify suitable candidates efficiently and consistently. This fragmentation slows down the recruitment process and increases the workload on HR teams.

2.2 Critique of the Existing System and Proposed Solution

2.2.1 Critique of the Existing System

Several limitations characterize the existing recruitment approach :

- Lack of centralization : Candidate data and job offers are spread across multiple platforms.

- Time-consuming CV screening : Manual sorting of resumes increases processing time and recruiter workload.
- Reduced decision accuracy : Comparing candidates becomes difficult without structured evaluation tools.
- Limited process visibility : Recruiters struggle to track application status and recruitment stages effectively.

These limitations reduce the overall efficiency of recruitment activities and prevent recruiters from focusing on strategic HR tasks.

2.2.2 Proposed Solution

To address these limitations, this project proposes the design and development of an intelligent recruitment portal for Sopra HR Software. The portal aims to digitalize and automate the recruitment process by integrating the following features :

- Centralized management of job offers and applications
- Automated CV collection and structured candidate profiles
- Support for CV screening and candidate shortlisting
- Tools to assist recruiters in evaluating candidates and making decisions
- A unified platform facilitating interaction between recruiters and applicants

By adopting an intelligent recruitment portal, Sopra HR Software can streamline hiring workflows, reduce recruitment time, and enhance the effectiveness and reliability of the recruitment process. This solution aligns with best practices in electronic recruitment systems, which have proven effective in simplifying hiring and improving organizational performance.

3. Development Methodology Choice

During the development of an information system, it is essential to adopt a clear, structured methodology to guide the project through its different stages. A development methodology

defines the processes and steps required to organize the work, ensure progress at each phase, and align the system being developed with user needs. For this project, the Agile methodology was chosen, as it provides flexibility, encourages continuous improvement, and supports incremental delivery throughout the development cycle.

3.1 Agile Method

Agile methodology is widely adopted today within IT service companies and web development agencies, as it is based on an iterative and incremental development cycle that promotes strong collaboration among team members. This approach aims to deliver a high-quality final product while continuously adapting to evolving client needs. The choice of the Agile methodology for this project is motivated by several factors :

- The functional scope of the project requires development across multiple iterations to validate each version of the system progressively.
- A strong level of interaction with stakeholders is essential to ensure better alignment with user requirements and to improve overall product quality.
- Enhanced and continuous communication within the project team fosters rigor, responsiveness, and efficiency throughout the development process.

In the following section, we focus in greater detail on the specific Agile approach adopted in this project to present the principles of incremental development and system evolution more concretely.

3.2 Scrum Methodology

Scrum is widely recognized as a project management framework that facilitates the efficient development of complex systems. It relies on an iterative and incremental approach, organized into short, time-boxed cycles known as sprints, typically lasting from a few days up to one month. Each sprint is guided by a clearly defined objective and results in the delivery of a potentially usable product increment.

Agile methodologies, particularly Scrum, are built upon four fundamental values :

- Individuals and interactions
- A functional and operational product
- Customer collaboration
- Responsiveness to change

This framework allows teams to adapt effectively to evolving requirements while promoting continuous improvement and maintaining high quality throughout the development process.

SCRUM Actors

Scrum is built around three main roles, each with clearly defined responsibilities :

- **Product Owner** : The Product Owner is responsible for defining the product vision and representing stakeholder needs. This role involves identifying project requirements, translating them into tasks, and prioritizing the product backlog to maximize value delivery.
- **Scrum Master** : The Scrum Master ensures that the Scrum framework is properly understood and applied by all participants. This role facilitates communication between the Product Owner and the development team, removes obstacles, and ensures that the project progresses efficiently according to Agile principles.
- **Development Team** : The development team is composed of 3 to 9 members who collaborate daily to design, develop, and deliver the expected system features at the end of each sprint.

Role Distribution in This Project

In the context of this project, the Scrum roles are assigned as follows :

- Product Owner : Mr. Ramzi BEN CHAMEKH
- Scrum Master : Ms. Nada BOUHLEL
- Development Team : Mayssem ESSMAILI and Bechir JABLI

This role distribution ensures effective coordination, clear responsibility sharing, and close alignment between project objectives and implementation, thereby supporting the successful development of the intelligent recruitment portal for Sopra HR Software.

3.3 Product Backlog

The Product Backlog is a structured list that contains all the features, functionalities, and requirements expected from the product. Scrum does not impose a specific format or naming convention for backlog items; however, they are mostly expressed as user stories or use cases. In a Product Backlog, these items are organized according to their potential implementation order, making prioritization a key aspect of iterative development. The Product Backlog serves as a guide for the development team and remains flexible, evolving throughout the project as requirements change. Its creation and ongoing refinement are collaborative activities involving both the Product Owner and the development team.

Each user story is characterized by several key attributes that facilitate planning and implementation :

- Priority : The level of importance assigned by the Product Owner reflects the business value and urgency of the feature.
- Initial Estimation : A preliminary evaluation of the effort required to implement the user story in relation to others, generally expressed in terms of time, such as days.

Sprint Backlog

Commonly referred to as the Sprint Backlog, this artifact represents the list of work that the team commits to completing during a sprint. It is often broken down into a set of more detailed and operational tasks derived from the selected user stories. The Sprint Backlog was initially created during the Sprint Planning Meeting, but it remains flexible and is refined by the development team as requirements evolve or new constraints arise. Each task within the Sprint Backlog is monitored during the sprint and reflects the remaining effort needed to achieve the sprint objectives.

4. Technology Choices

4.1 Software Tools

For the development of the *Sopra Recruit* application, several software tools were selected to meet the project's specific requirements. The following section describes the main tools used during the development process.

StarUML : StarUML is a UML modeling tool dedicated to software engineering and system design. It was used in this project to create UML diagrams, such as use case diagrams and sequence diagrams, allowing a structured and formal representation of the system's functionalities and interactions.



FIGURE 2 – StarUML Logo

Visual Studio Code : The source code editor used during the internship for software development. Developed by Microsoft, it supports multiple programming languages and offers key features such as syntax highlighting, code auto-completion, debugging tools, and Git integration.



FIGURE 3 – Visual Studio Code Logo

IntelliJ IDEA : IntelliJ IDEA is the integrated development environment (IDE) used during the internship for the development of the backend application based on Java and Spring Boot. Developed by JetBrains, it is particularly well suited for enterprise Java development, offering advanced features such as intelligent code completion, efficient project structure management, built-in debugging, and seamless integration with Spring frameworks.



FIGURE 4 – IntelliJ IDEA Logo

Java Development Kit (JDK) : The Java Development Kit is a fundamental software package used during the internship for developing and executing Java-based backend applications. It provides essential tools such as the Java compiler, runtime environment, and standard libraries required for application development.



FIGURE 5 – Java Development Kit (JDK) Logo

Spring Boot : Spring Boot is a Java-based framework used during the internship for the development of the backend application. It simplifies the creation of production-ready Spring applications by providing automatic configuration, embedded servers, and built-in support for RESTful services.



FIGURE 6 – Spring Boot Logo

Apache Maven : Apache Maven is a build automation and dependency management tool used during the internship for the development of the Spring Boot backend application. It simplifies project configuration by managing dependencies, build processes, and project structure through a centralized configuration file.

PostgreSQL : PostgreSQL is an open-source relational database management system used during the internship to store and manage the backend application data. It provides robust



FIGURE 7 – Apache Maven Logo

support for structured data, transactions, and data integrity through advanced querying and constraint mechanisms.



FIGURE 8 – PostgreSQL Logo

Postman : Postman is an API testing and development tool used during the internship to test and validate the backend RESTful web services developed with Spring Boot. It enables the sending of HTTP requests and the analysis of server responses.

Git and GitHub : Git is a distributed version control system used during the internship to manage and track changes in the project source code. GitHub served as a remote repository hosting platform, enabling source code storage, version history management, and collaboration.

React.js : React.js is a JavaScript library used during the internship for building the frontend of the web application. It enables the development of reusable and modular user interface components through component-based architecture.

Vite : Vite is a modern frontend build tool used to initialize and run the React application during the internship. It provides fast development startup times and efficient module bundling.

Tailwind CSS : Tailwind CSS is a utility-first CSS framework used during the internship to style the frontend application. It provides pre-defined utility classes that allow rapid and consistent user interface design directly within component files.

Conclusion

In this first chapter, we presented the general context of the internship by introducing the host organization and defining the objectives of the project. The chapter outlined the problem to be addressed and justified the need for a well-structured web application to meet functional and technical requirements. In addition, an overview of the software tools, backend and frontend technologies used throughout the project was provided, highlighting their roles in the development process. This chapter therefore establishes the methodological and technical foundation upon which the subsequent design, implementation, and evaluation phases of the project are built.